

Superior stock trading performance is achieved with proprietary LLMs as agent backbones

Compared to agents employing open-source or financial-domain-specific fine-tuned LLMs, those using the three proprietary LLMs demonstrated significantly higher and more consistent average CR and SR, as shown in Figure 3a. Despite being fine-tuned with extensive financial contexts, domain-specific LLMs did not provide a decisive advantage in sequential stock trading decision-making tasks. This may be attributed to their primary training for other functions, such as long financial report analysis exemplified by Palmyra-Fin-70B, rather than decision-making.

Model parameter size increment enhances agent financial decision-making quality and robustness. In the category of open-source LLMs, those exceeding 67B parameters displayed superior CRs and SRs, along with markedly less variance within their category, as illustrated in Figure 3b and Table 2. This underscores the prevailing belief that the reasoning capabilities of LLMs are proportionate to their parameter size, which holds also true in stock trading, which is a sequential decision-making task in an open-ended, volatile environment by nature. **Proprietary models exhibit significantly stronger decision-making capabilities compared to even the largest open-source LLMs under complex, mixed market conditions,** though this advantage is less evident in relatively monotone market environments. During the test phase, primarily influenced by the range of open-source data collected, TSLA and NIO exhibited volatility with mixed upward and downward stock price trends, whereas the other five stocks generally showed bullish trends. The investment signals derived from such complex markets tend to be noisy or delayed, as illustrated in Appendix C. We observed that proprietary models possess a superior ability to manage these challenging conditions and consistently deliver better performance outcomes than large-sized open-source LLMs. Their reasoning capability enables them to effectively utilize other decision-relevant information, such as historical momentum, current holdings, and, most critically, self-reflection outcomes from the agents, thereby facilitating more accurate decisions.

4.3 Result 2 & 3: Cryptocurrency Trading and ETF Trading

In the test phases of both cryptocurrency and ETF trading tasks, market trends are mixed. Notably,

Table 3: Performance of cryptocurrency trading with different LLMs as backbone models across Bitcoin (BTC) and Ethereum (ETH).

Model	BTC				ETH			
	CR $\uparrow$	SR $\uparrow$	AV $\downarrow$	MDD $\downarrow$	CR $\uparrow$	SR $\uparrow$	AV $\downarrow$	MDD $\downarrow$
Buy & Hold	21.821	0.683	37.426	20.796	4.528	0.146	41.817	29.889
<i>Financial Domain Models</i>								
Palmyra-Fin-70B	-20.812	-1.212	20.036	27.782	4.795	0.240	26.924	16.405
<i>Proprietary Models</i>								
GPT-o1-preview	34.060	1.114	35.846	17.075	2.496	0.085	39.641	27.692
GPT-4	22.396	0.828	31.699	17.206	1.516	0.051	39.812	32.541
GPT-4o	14.330	0.532	31.304	17.278	4.666	0.190	33.051	22.539
Average	23.595	0.825	32.950	17.186	2.893	0.109	37.501	27.591
<i>Open-Source Models</i>								
Qwen2.5-72B-Instruct	0.549	0.325	1.979	0.897	11.984	0.584	18.554	27.642
Llama-3.1-70B-Instruct	20.440	0.758	31.604	17.813	-11.888	-0.410	39.047	36.416
DeepSeek-67B-Chat	28.307	0.891	37.219	17.944	9.480	0.309	41.369	26.261
Yi-1.5-34B-Chat	13.620	0.434	36.778	22.790	6.325	0.227	37.503	25.707
Qwen2.5-32B-Instruc	11.566	0.869	15.608	7.984	2.823	0.194	19.571	7.883
DeepSeek-V2-Lite (15.7B)	4.804	0.153	36.846	20.562	-9.504	-0.311	41.199	21.270
Yi-1.5-9B-Chat	7.953	0.253	36.799	26.545	-3.684	-0.119	41.818	35.417
Llama-3.1-8B-Instruct	20.521	0.646	37.240	21.104	4.939	0.163	40.928	29.466
Qwen-2.5-Instruct-7B	19.477	0.612	37.289	20.796	-1.339	-0.075	24.124	-16.053
Average	14.137	0.549	30.151	17.382	1.015	0.062	33.790	21.557

the cryptocurrency task shows significantly smaller price fluctuations compared to the ETF task. We outline the key features of using an LLM-agent to make financial decisions across these two distinct markets as follows:

Large-sized open-source models and proprietary models are needed to effectively capture trading signals of cryptocurrency markets, which are highly sensitive to news and financial sentiment. As shown in Table 3, using mid-sized and small-sized open-source models as the decision-making agent backbone generally results in weaker performance than the market baseline with respect to CR and SR.

ETF investment requires proprietary models enriched with extensive pre-trained knowledge to serve as the agent’s “brain” and provide robust reasoning support. As shown in Table 4, proprietary models significantly outperform open-source and financial domain-specific models in this task. This advantage arises from the complexity of ETF trading, which necessitates interpreting actionable signals across diverse sectors, demanding more strategic, long-term decisions grounded in deep comprehension and reflection anchored by rich pre-contexts.

4.4 Discussion

Combining all the experimental results, we find that the performance of different LLM varies significantly in stock, cryptocurrency, and ETF trading. This variation not only reflects the inherent complexity of financial markets, but also highlights the importance of model selection or fine-tuning. For instance, proprietary LLM generally exhibit be

Table 4: Performance of ETF trading with different LLMs as backbone models.

ETF	CR $\uparrow$	SR $\uparrow$	AV $\downarrow$	MDD $\downarrow$
Buy & Hold	2.069	0.06	46.645	35.746
<i>Financial Domain Models</i>				
Palmyra-Fin-70B	24.759	1.152	30.419	8.203
<i>Proprietary Models</i>				
GPT-o1-preview	21.224	0.849	43.766	20.054
GPT-4	2.807	0.110	44.679	37.785
GPT-4o	12.292	0.377	46.150	32.678
Average	12.108	0.445	44.865	30.172
<i>Open-Source Models</i>				
Qwen2.5-72B-Instruct	4.507	0.227	28.090	8.580
Llama-3.1-70B-Instruct	9.895	0.464	30.184	12.759
Yi-1.5-34B-Chat	4.996	0.322	21.986	12.858
Qwen2.5-32B-Instruct	19.617	0.955	29.070	7.496
DeepSeek-V2-Lite (15.7B)	1.389	0.063	31.371	31.831
Yi-1.5-9B-Chat	-4.657	-0.228	28.907	15.545
Llama-3.1-8B-Instruct	11.239	0.475	33.480	15.587
Qwen-2.5-Instruct-7B	-0.384	-0.020	27.596	14.059
Average	5.825	0.282	28.835	14.839

performance in stock trading due to their strong training on various financial datasets, while open-source models struggle to achieve these results, especially in more volatile environments such as cryptocurrency trading. In addition, the effectiveness of LLM-based agents depends heavily on their ability to adapt to market fluctuations. Agents that incorporate advanced memory systems and dynamic risk assessment capabilities are better able to cope with complex market situations, highlighting the value of the complex architectural features of LLM-based agent framework in financial decision-making tasks.

5 Related Work

5.1 LLM for Financial Domain

The rapid development of general-domain language models (LMs) has stimulated the exploration of financial LMs, such as pre-trained LMs: FINBERT (Liu et al., 2021; Yang et al., 2020; Araci, 2019; Huang et al., 2023), FINBERT-MRC (Zhang and Zhang, 2023), FLANG (Shah et al., 2022), and several financial LLMs: FINGPT (Liu et al., 2023), FINMA (Xie et al., 2023), INVESTLM (Yang et al., 2023), BloombergGPT (Wu et al., 2023), which leverage extensive training on diverse financial datasets (e.g. stock price data, financial news and analyst reports) and adapt the capabilities of LMs to the unique needs of financial applications. Concurrently, the advancement of LLMs has significantly enhanced the development of language-based agent frameworks in the finan-

cial sector, such as FINMEM (Yu et al., 2024a), FINAGENT (Zhang et al., 2024a) and FINROBOT (Yang et al., 2024), characterized by their adaptability and openness. However, variations in framework design, task scope, and data types present challenges in uniformly evaluating the efficacy of LLM agents in financial scenarios.

5.2 Financial LLM Benchmarks

In the realm of financial LLMs, several benchmarks have been developed: FLUE (Shah et al., 2022) introduces the first comprehensive benchmark with five financial NLP tasks, including sentiment analysis, headline classification, named entity recognition, structure boundary detection, and question answering. Pixiu (Xie et al., 2023) expands this benchmark to include financial document understanding and classification tasks, incorporating multimodal datasets. FinBen (Xie et al., 2024) encompasses 36 datasets covering 24 financial tasks. Despite these advancements, there remains a notable gap in benchmarks specifically designed for LLM-based agent applications within the financial sector.

6 Conclusion

INVESTORBENCH offers the community two distinct modes of engagement. The first mode allows participants to integrate their fine-tuned LLMs into the INVESTORBENCH’s agent framework to undertake financial decision-making tasks. This setup enables them to benchmark the performance of their models against those previously experimented with by our work. The second mode permits users to directly incorporate the environment and evaluation metrics of INVESTORBENCH into their own designed agents, facilitating a comparative analysis of their agent design’s effectiveness. This dual approach provides a flexible framework for testing and enhancing financial decision-making strategies within the INVESTORBENCH ecosystem.

Future research efforts will expand the benchmark by incorporating additional information modalities, such as audio (e.g., earnings call recordings) and graphs (e.g., K-lines, trade charts), to explore whether these data types can enhance decision-making quality. The foundational agent framework of INVESTORBENCH is designed to seamlessly accommodate these modalities, ensuring that the extended benchmark remains easy to use and scalable.

Limitation

First, INVESTORBENCH is currently focusing on single-asset financial decision-making task, without addressing multi-asset tasks such as portfolio management. Second, copyright restrictions on financial domain data may compromise the quality of the datasets we create, potentially limiting the assessment of model performance.

Ethical Statement

The authors take full responsibility for the development of INVESTORBENCH, ensuring that the publicly available part in dataset does not contain personal information, and conform to established ethical guidelines. The data are shared under the MIT license, requiring users to adhere to its terms. INVESTORBENCH is intended for academic and educational purposes only and is not a substitute for professional advice. While efforts have been made to ensure its accuracy, the authors and their institutions disclaim liability for any outcomes arising from its use. Users agree to take responsibility for ethical and lawful use and to indemnify the authors and their affiliates against any claims or damages resulting from reliance on this Material.

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